

Q2(iii)

```
def euler_method(f, x0, t0, dt, num_step):

    t = [t0]
    x = [x0]

    t_1 = t0
    x_1 = x0

    for _ in range(num_step):
        dx_dt = f(t_1, x_1)
        x_2 = x_1 + dt*(dx_dt)
        t_2 = t_1 + dt

        t.append(t_2)
        x.append(x_2)

        t_1 = t_2
        x_1 = x_2

    return t,x

def bac_growth(t, x):
    a = 0.1
    b = 0.001
    return a*x - b*x**2

x0 = 2000
t0 = 0
total_time = 4
num_step = 500

dt = total_time/ num_step
time_values, population_values = euler_method(bac_growth, x0, t0, dt,
num_step)

final_population_euler = population_values[-1]
final_time_euler = time_values[-1]

print(f"Analytical solution for x(4) ≈ {275.31:.2f}")
print(f"Euler's Method result at t={final_time_euler:.2f} hours:
{final_population_euler:.2f} bacteria")

Analytical solution for x(4) ≈ 275.31
Euler's Method result at t=4.00 hours: 274.49 bacteria
```